UNIT III

Chapter 7

Body Fluids and Circulation

Chapter Outline

7.1 Body fluids
7.2 Blood vessels – Arteries, Veins and capillaries
7.3 Circulatory path ways
7.4 Human circulatory system
7.5 Double circulation
7.6 Regulation of cardiac activity
7.7 Disorders of the circulatory system
7.8 Cardio pulmonary resuscitation (CPR)



Stroke volume is dependent on venous return

Learning Objectives:

- *Understands the importance of body fluids.*
- *Identifies and describes the blood cells, different types of blood groups and blood coagulating factors.*
- *Differentiate the blood vessels and its properties*
- *Understands the human circulatory system.*
- *Understands the cardiac cycle and relate with the peaks of ECG.*
- *Identifies the disorders of circulatory system.*

Animals particularly larger animals like mammals, are more active. They depend on locomotion to find food which is an energy consuming process. Nervous system is required to coordinate activities by sending nerve impulses that involves energy. All living cells have to be supplied with nutrients, oxygen and other substances and have to remove CO_2 and waste products from them. It is therefore essential to have efficient mechanisms for transport of these substances to and from the cells. Different groups of animals have evolved different methods of transport. Very small organisms like the sponges and coelenterates lack a circulatory system. Water from their surroundings enters their body cavity to facilitate the cells to exchange substances by diffusion. More complex organisms use special fluids and well organized transport systems within their body to transport such materials by **bulk flow** or connective transport with pumps. The phenomenon of bulk flow is fundamental to many physiological processes like respiration, digestion and excretion. The bulk flow of fluids can transport substances to long distances faster than by diffusion. The

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